PH4102: Introductory Astrophysics

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Spectral types of star:

Star's spectra were known to the astronomers even before a proper theoretical understanding of those the spectra were available. As early as 1817, Joseph Fraunhofer had determined that different stars have different spectra.

Astronomers started devloping catalogues of the stars based on the *hydrogen absorption lines* in the observed spectra. Such works were pioneered at Havard by Edward C. Pickering (1846–1919) and his assistant Williamina P. Fleming (1857–1911) in the 1890s and latter improved by their successors at Havard.

After subsequent modifications, the presently acceptable Harvard classification scheme is "O B A F G K M" and it represents a *temperature sequence*, running from the hottest blue O stars to the coolest red M stars. Decimal subdivisions (e.g., A0–A9) is are also included - these labels distinguish the stars within the spectral subdivisions. Stars nearer the beginning of this sequence are referred to as *early-type stars*, and those closer to the end are called *late-type stars*. So astronomers may speak of a K0 star as an "early K star" or refer to a B9 star as a "late B star". Sun's type is G2.

Annie Jump Cannon classified some 200,000 spectra between 1911 and 1914, and the results were collected into the **Henry Draper Catalogue**. Even today many stars are referred by their HD numbers (e.g., Betelgeuse is HD 39801).

Very recently two more classes L and T are added to to the above "O B A …" list.

Theory for the spectral features of stars were not known until the development of quantum mechanics and subsequent development of Boltzmann equations (for excitation), Saha ionization equations, quamtum statitics etc.

The formation of spectral lines (emission or absorption) is quite complicated and depends non-linearly on above theories.

Presently, we know that the spectrum (except the lines) of a star can be fitted with a black body spectrum and the effective temperature (T_{eff}) of the star can be found out. So, the different classes of spectral classification represent different surface temperatuers of the stars.

TABLE 1 Harvard Spectral Classification.

Spectral Type	Characteristics
O	Hottest blue-white stars with few lines Strong He II absorption (sometimes emission) lines. He I absorption lines becoming stronger.
B	Hot blue-white He I absorption lines strongest at B2. H I (Balmer) absorption lines becoming stronger.
A	White Balmer absorption lines strongest at A0, becoming weaker later. Ca II absorption lines becoming stronger.
F	Yellow-white Ca II lines continue to strengthen as Balmer lines continue to weaken. Neutral metal absorption lines (Fe I, Cr I).
G	Yellow Solar-type spectra. Ca II lines continue becoming stronger. Fe I, other neutral metal lines becoming stronger.
K	Cool orange Ca II H and K lines strongest at K0, becoming weaker later. Spectra dominated by metal absorption lines.
M	Cool red Spectra dominated by molecular absorption bands, especially titanium oxide (TiO) and vanadium oxide (VO). Neutral metal absorption lines remain strong.
L	Very cool, dark red Stronger in infrared than visible. Strong molecular absorption bands of metal hydrides (CrH, FeH), water (H_2O), carbon monoxide (CO), and alkali metals (Na, K, Rb, Cs). TiO and VO are weakening.
T	Coolest, Infrared Strong methane (CH_4) bands but weakening CO bands.

Different stars have different composition as well as different surface temperature.

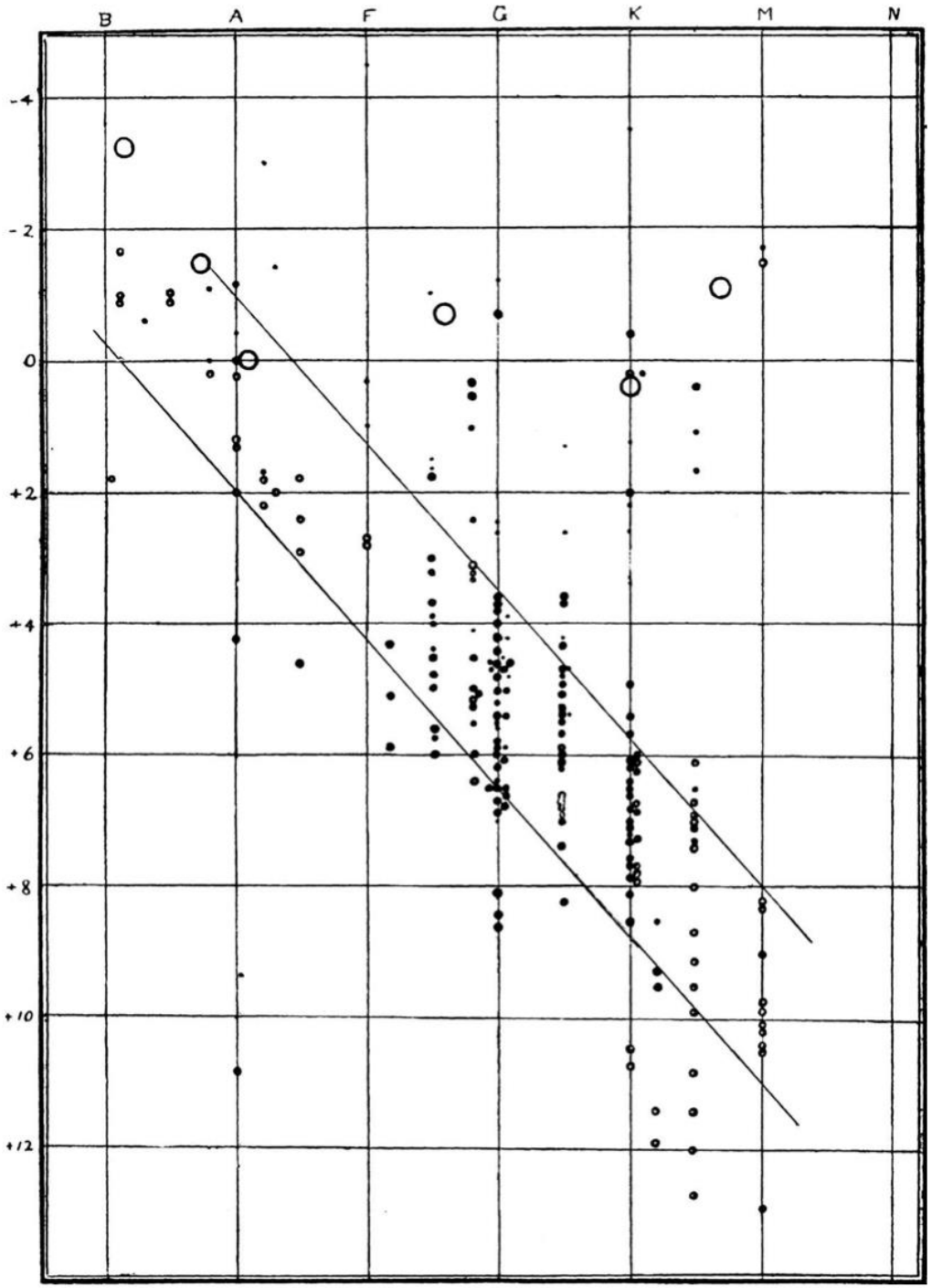
Hertzsprung–Russell (H–R) diagram

Early in the twentieth century, as astronomers accumulated data for an increasingly large sample of stars, they became aware of the wide range of stellar luminosities and absolute magnitudes.

A Danish engineer and amateur astronomer, E. Hertzsprung (1873–1967), analyzed stars whose absolute magnitudes and spectral types had been accurately determined. He found expected correlation between these two for most stars except stars of type G or later. He found that stars of type G or later had a range of magnitudes, despite having the same spectral classification. Hertzsprung termed the brighter stars **giants**.

Around same time, at Princeton University, Henry N. Russell (1877–1957) independently came to the same conclusions as Hertzsprung. Russell used the same term, *giant*, to describe the luminous stars of late spectral type and the term **dwarf** stars for their dim counterparts.

In their plot, they showed the correlation for more than 200 stars. Such plots for are named **H-R** diagram following their names.



Plot of spectral class vs. absolute magnitude

More than 200 stars were plotted, most within a band reaching from the upper left-hand corner, home of the hot, bright O stars, to the lower right-hand corner, where the cool, dim M stars reside. This band, called the **main sequence**, contains between 80% and 90% of all stars in the H–R diagram. In the upper right-hand corner are the giant stars. A single **white dwarf**, 40 Eridani B, sits at the lower left.

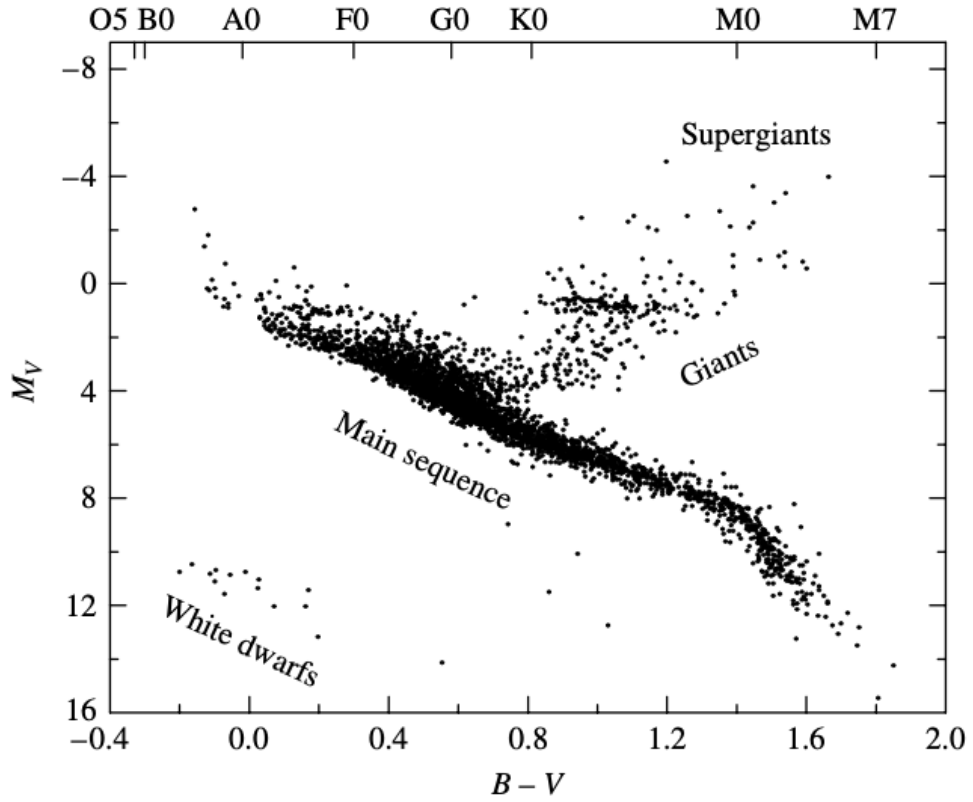
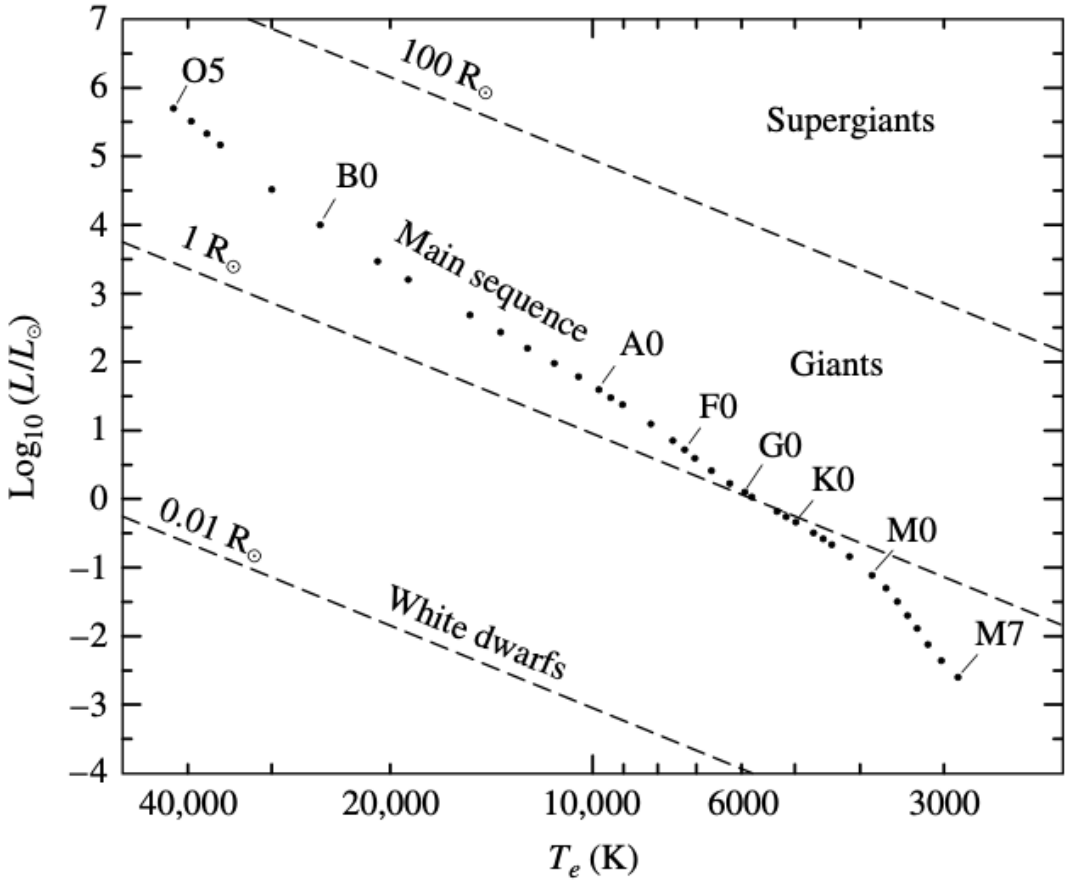


FIGURE 13 An observer's H–R diagram. The data are from the Hipparcos catalog. More than 3700 stars are included here with parallax measurements determined to better than 20%. (Data courtesy of the European Space Agency.)

Recent version of an observational H–R diagram.

Absolute visual magnitude vs. color index and spectral type.



This is known as theorists' version of H-R diagram.
Based on the average properties of main-sequence stars, this diagram has a theorist's orientation: The luminosity and effective temperature are plotted for each star, rather than the observationally determined quantities of absolute magnitude and color index or spectral type.

The radius of a star can be easily determined from its position on the H–R diagram. The Stefan–Boltzmann law shows that if two stars have the same surface temperature, but one star is 100 times more luminous than the other, then the radius of the more luminous star is √100 = 10 times larger. On a logarithmically plotted H–R diagram, the locations of stars having the same radii fall along diagonal lines that run roughly parallel to the main sequence.

The main-sequence stars show some variation in their sizes, ranging from roughly 20 $R_\odot$ at the extreme upper left end of the main sequence down to 0.1 $R_\odot$ at the lower right end.

The giant stars fall between roughly 10 $R_\odot$ and 100 $R_\odot$.

The supergiant stars are even larger. Betelgeuse, a pulsating variable star, contracts and expands roughly between 700 and 1000 times the radius of the Sun with a period of approximately 2070 days.

The existence of such a simple relation between luminosity and temperature for main sequence stars is a valuable clue that the position of a star on the main sequence is governed by a single factor. This factor is the star's mass.

The most massive O stars are observed to have masses of 60 $M_\odot$, and the lower end of the main sequence is bounded by M stars having at least 0.08 $M_\odot$